Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TECHNICAL CATALOGUE · HAND-DRIVEN · METAL BODY

CIE/444

Hand-driven insulation tester in anti-magnetic aluminium

Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant equipment, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, electrical components, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.



CIE/444 · ILLUSTRATIVE IMAGE

27	6	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

100 V · 250 V · 500 V · 1000 V · 2500 V · 5000 V

01 HIGHLIGHTS

- ✓ Anti-magnetic aluminium body
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Protected metallic flap over the scale window
- ✓ Long life and strength against vibration and impact
- ✓ Guard terminal and externally accessible infinity adjuster on ranges of 1000 MΩ and above
- ✓ Silent generator
- ✓ Collapsible driven handle, folds back flush into the casing
- ✓ Wide anti-parallax mirror scale (not provided on triple-range and above)
- ✓ Highly sensitive cross-coil indicator
- ✓ Accurate and quick response: 3 seconds
- ✓ Conforms to IS 2992:1987

02 TYPICAL APPLICATIONS

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and communication cable insulation testing
- Transformer and insulator testing; industrial installation checks
- Locating intermittent shorts, defective connections, and developing insulation failures

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03 PRINCIPLE AND CONSTRUCTION

PRINCIPLE

A hand-driven clutch-type A.C. generator and a highly sensitive cross-coil indicator. The generated A.C. voltage is converted to D.C. voltage by silicon diodes.

CONSTRUCTION

Housed in a sturdy, elegant case of anti-magnetic cast aluminium with a metallic carrying handle. The instrument is dust-free: rubber shoes hold the surface and give it stability, and a protected metallic flap covers the glass window of the scale.

04 SALIENT FEATURES

Silent generator

The A.C. generator runs extremely silently and smoothly because of the excellent dynamic balance of its rotating parts, and is free from mechanical noise.

Centrifugal governor

A high-grade centrifugal governor holds the voltage constant for accurate measurement. Variation in crank speed does not affect the reading, as a cut-off clutch disengages the drive beyond the rated speed.

Collapsible handle

The generator driven handle folds back flush into the casing when the tester is not in use.

Quick response

The response time is 3 seconds, when a resistance equal to the central scale mark is suddenly inserted between the measurement terminals, as per IS 2992:1987.

05 PERFORMANCE AND TEST SPECIFICATION

Standard	IS 2992:1987
Indicator	High-sensitivity cross-coil movement; wide anti-parallax mirror scale (not provided on triple-range and above instruments)
Scale length	90 mm (approx.)
Accuracy	Conforms to IS 2992:1987
Generator speed	160 r.p.m. or more; the centrifugal governor holds the output voltage constant
Response time	3 seconds
High voltage test	Withstands minimum 2000 V or (rated voltage + 1000) V A.C. (rms) between the electrical circuit and the handle for one minute
Insulation resistance of the instrument	Better than 50 MΩ at 500 V between the electrical circuit and the containing case

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06 CONSTRUCTION AND PHYSICAL SPECIFICATION

Guard terminal and infinity adjuster	Provided on instruments with an insulation range of 1000 MΩ and above
Body	Anti-magnetic cast aluminium case with metallic carrying handle and protected metallic flap over the scale window
Weight	2.5 kg (approx.)
Dimensions	110 × 128 × 195 mm (approx.)

07 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
100 V	0–20 · 0–50 · 0–100
250 V	0–50 · 0–100 · 0–250
500 V	0–100 · 0–200 · 0–500 · 0–1000 · 0–2000
1000 V	0–100 · 0–200 · 0–500 · 0–1000 · 0–2000
2500 V	0–1000 · 0–2000 · 0–2500 · 0–5000 · 0–10,000 · 0–20,000 · 50–50,000
5000 V	0–5000 · 0–10,000 · 0–20,000 · 50–50,000

The 100,000 MΩ range at 5000 V is not offered on the metal-body CIE/444 (available on the wooden-body CIE/666).

08 COMBINING VOLTAGES AND RANGES

Dual voltage and dual insulation range

One instrument with two rated voltages, each with its own insulation range, selected by a switch. The selected range must be within the rated resistance listed for that voltage (voltage to measurable insulation range ratio up to 1:2 only; for any other ratio, please check with us).

Single voltage, dual insulation range

One rated voltage with two insulation ranges. The selected ranges must be within the rated resistance listed for that voltage.

Continuity range (500 V and 1000 V only)

Available as an extra on single-voltage, single-range instruments, or on dual-voltage, dual-range instruments.

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09 MULTIRANGE SERIES: CIE/444/MR

SERIES	RATED VOLTAGE / INSULATION RANGE (MΩ)	CONTINUITY
MR-1	1000 V / 0–2000 500 V / 0–1000 250 V / 0–500	0–20 kΩ
MR-2	1000 V / 0–200 500 V / 0–100 250 V / 0–50	0–1 kΩ
MR-3	500 V / 0–1000 250 V / 0–500 100 V / 0–200	0–5 kΩ
MR-4	500 V / 0–100 250 V / 0–50 100 V / 0–20	0–1 kΩ

Multirange instruments are provided with a range selector switch, so one instrument covers every voltage and range listed in its series. Anti-parallax mirror scale is not provided on triple-range and above instruments. Any other range can be manufactured against specific requirements.

10 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case (metal body)
- Testing leads, 1 m or 2 m (up to 1000 V)
- H.V. testing leads, 2 m (general / special)

11 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/444, 500 V / 0–2000 MΩ**. Any range not listed can be manufactured against specific requirements.

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12 CIE HAND-DRIVEN AND MOTOR-OPERATED INSULATION TESTERS COMPARED

	CIE/444	CIE/666	CIE/555	CIE/777	CIE/777 HM
Drive	**Hand-driven**	Hand-driven	Hand-driven	Mains motor, 230 V, 50 Hz	Mains motor, 230 V, 50 Hz, or hand cranking
Body	**Anti-magnetic cast aluminium**	Seasoned wood, sunmica finish	Moulded bakelite	Seasoned wooden cabinet, sunmica finish	Seasoned wooden cabinet, sunmica finish
Rated voltages (V)	**100, 250, 500, 1000, 2500, 5000**	100, 250, 500, 1000, 2500, 5000, 10,000	100, 250, 500, 1000	500, 1000, 2500, 5000, 10,000	500, 1000, 2500, 5000, 10,000
Standard range combinations	**27**	32	11	14	14
Multirange series	**MR-1 to MR-4**	MR-1 to MR-8	Single range only	MR-A to MR-E	MR-A to MR-E
Scale length	**90 mm**	100 mm	75 mm	100 mm	100 mm
Response time	**3 seconds**	3 seconds	3 seconds	3 seconds	3 seconds
Weight (approx.)	**2.5 kg**	4.1 kg	1.4 kg	9 kg (incl. top cover)	12 kg (incl. top cover)
Conforms to	**IS 2992:1987**	IS 2992:1987	IS 2992:1987	IS 2992:1987	IS 2992:1987

Rated voltages, standard ranges and multirange series as listed in this catalogue series. Weights and scale lengths are approximate.

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